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Executive Summary

This report analyzes over 3,200 cleaned YouTube comments to measure how consumer sentiment shifted after Google discontinued three products: Google Stadia, Google Glass, and Google+. Using general sentiment analysis, TF-IDF keyword analysis, keyword co-occurrence networks, and aspect-based sentiment analysis (ABSA), we found three distinct post-discontinuation sentiment trajectories.

Google Glass experienced a rare positive sentiment reversal, evolving from a criticized product into a nostalgic "pioneer" with strong brand halo potential. Google+ remains persistently negative, with resentment concentrated on Google's corporate decision-making rather than the product itself. Google Stadia presents the highest reputational risk, where praise for excellent technology has been overpowered by a deep sense of betrayal over its shutdown.

We recommend that any organization with frequent product turnover institutionalize pre- and post-mortem ABSA to quantify reputational injury at the feature level, enabling data-driven communication strategies that rebuild trust where it is most broken.

Overview: This report analyzes how public sentiment toward three discontinued Google products changed over time, using YouTube comment data scraped before and after each product's shutdown. The three products are Google Stadia, Google Glass, and Google+.

The core research questions: *Did consumer sentiment shift significantly after Google announced the discontinuation of these products?*

What factors were identified as the core drivers of discontinuation by sentiment analysis?

What recommendations do we have for other organizations developing & selling similar products?

Product Descriptions:

Google Stadia: Google Stadia was a cloud-based gaming platform released on November 19, 2019, and discontinued on January 18, 2023. It allowed users to play high-quality video games without a traditional console by streaming gameplay directly from Google's servers to devices such as TVs, laptops, and smartphones. Key features included instant access to games without downloads, cross-device play, integration with YouTube for streaming and sharing, and a dedicated Wi-Fi-enabled controller. In non-technical terms, Stadia functioned like a "Netflix for gaming," enabling users to play games over the internet rather than relying on physical hardware. Despite its innovative approach and strong technical performance, the platform faced challenges related to its game library, value proposition, and user adoption.

Google+ was a social networking platform launched on June 28, 2011, and discontinued for consumers on April 2, 2019. It was designed to compete with other social media platforms by allowing users to connect, share content, and communicate within organized groups. Its core features included "Circles," which enabled users to categorize contacts for more controlled sharing; "Streams," which functioned as a content feed; "Communities" for group discussions; and "Hangouts" for video chatting and messaging. It was also deeply integrated with other Google services such as YouTube and Gmail. In simple terms, Google+ was Google's attempt to create a more organized and customizable social media experience. However, it struggled to achieve sustained user engagement and ultimately failed to build the strong network effects necessary for long-term success.

Google Glass was a wearable smart glasses device first released to developers and early adopters on April 15, 2013, with its consumer version discontinued on January 15, 2015 (though enterprise versions continued until March 2023). The device featured a small display positioned above the user's line of sight, enabling augmented reality experiences by overlaying digital information onto the real world. It included a built-in camera for photos and videos, voice-command functionality for hands-free operation, and the ability to display notifications such as messages, navigation, and weather updates. In non-technical terms, Google Glass functioned like a smartphone built into a pair of glasses, allowing users to access information and capture content without using their hands. While widely recognized as an innovative and forward-looking product, it faced challenges related to privacy concerns, usability, and mainstream adoption.

Data

YouTube comments were scraped for each product across two time windows — before discontinuation and after. After cleaning (removing duplicates, stripping URLs, filtering comments under 4 words), we ended up with approximately 3,248 usable comments across all three products.

Google Glass: Before discontinuation sentiment data was scraped from multiple videos that discussed the product. Two of the videos came from Google itself, on their Google Glass youtube account and their Google for Developers account. Each video has accumulated over 2 million views as of May 2026. The (April 2013 post date) Google Glass video titled Google Glass How-to: Getting Started has over 3,600 comments, and the (June 2013 post date) Google for Developers video titled Project Glass: Live Demo At Google I/O has over 1,800 comments as of May 2026. For both videos, most comments are concentrated within 1 - 2 years of the upload date. This means the majority of the comment data scraped from the videos are prior to the discontinuation of the “Explorer” edition of Google Glasses in January 2015, which was sold to individual consumers. Google would later sell an enterprise edition, starting in 2017, but this product line is not being analyzed in this report. The comments of three additional videos were used for sentiment analysis. These videos were made by popular content creators without official affiliations to Google. The table below shows overall engagement data for each video.

Pre-Discontinuation

Creator	Video Title	Post Date	Views	Comments
Smosh	Google Glass Sucks	April 19, 2013	32.7 Million	21,320
iJustine	Google Glass Review	June 28, 2013	3.3 Million	6,100
Marques Browlee	Google Glass Explorer Edition Explained	August 30, 2013	6.35 Million	14,800

The data scraping performed narrowed down the many thousands of comments to 684 pre-discontinuation data points.

Post-Discontinuation

Creator	Title	Post Date	Views	Comments
Dank Pods	Google Glass,	July 4, 2024	929,000	1,959

	but its 2024			
Heaven for Sale	Why Google Glass failed	March 13, 2023	17,110	34
Slidebean	What Happened to Google Glass	March 5, 2020	162,962	231
Surfshark Academy	Google Glass was cool	October 27, 2023	11,854	26
How It Happened	Why Google Glass Failed... and Why It's Coming Back	September 5, 2020	43,000	75

Post discontinuation data for Google Glass is derived from the videos detailed in the table above. It is not surprising that 5 or more years after the discontinuation of the product, there is far less comment data to scrape. Overall, there were 2,325 individual comments to scrape, which were narrowed down into 285 usable individual post discontinuation data points. Data was cleaned by removing duplicates, stripping URLs, and removing comments under 4 words.

Google Stadia: Same as Google Glass, pre and post discontinuation sentiment data was scraped from several videos discussing Google Stadia.

The pre-discontinuation data came from the videos in the table below.

Creator	Video Title	Post Date	Views	Comments
gameranx	Google Stadia - Before You Buy	November 19, 2019	1.44 Million	4,994
Austin Evans	I TRIED Google Stadia	November 22, 2019	2 Million	4,558
Unbox Therapy	Google Stadia Is Here. Is Xbox Doomed?	March 19, 2019	2.5 Million	10,351
Google	Stadia GDC 2019 Gaming	March 19, 2019	1.98 Million	4,247

Creator	Video Title	Post Date	Views	Comments
gameranx	Google Stadia - Before You Buy	November 19, 2019	1.44 Million	4,994
	Announcement			
TheRelaxingEnd	Google Stadia "Console" Unboxing - The Future of Gaming? (Gameplay Review + Controller)	November 21, 2019	4.936 Million	7,867

From these 32,017 comments, using the same means of data cleaning as before, we narrowed down the data into 594 individual pre-discontinuation data points for Google Stadia. Most of the comment data used was concentrated within 1-2 years of video upload dates. This means that the data does reflect pre-discontinuation sentiment as it falls in the period after the release of Stadia in 2019, but prior to its discontinuation in 2022.

Post-Discontinuation Data sources:

Creator	Video Title	Date Posted	Views	Comments
WatchMojo	Top 10 Reasons Why Google Stadia Failed	February 11, 2022	107,028	1,108
penguinz0	RIP Google Stadia	October 14, 2022	1.58 Million	3,272
Scott's Stash	Looking Back at Google Stadia	December 15, 2025	378,506	1,401

TripleJump	What Went Wrong With Google Stadia?	April 7, 2023	169,761	597
KnowledgeHusk	Why Did Stadia Fail?	October 17, 2022	256,725	753

Post-discontinuation data sources yielded far less comments (7,131 vs. 32,017), but still provided a quantity significant enough to be used for meaningful analysis. Staying consistent with the trend seen with Google Glass, there is far less engagement regarding a product post-discontinuation. That being said, when conducting data cleaning, the 7,131 comments were narrowed down to 609 post-discontinuation data points. This slightly exceeds the number of Stadia pre-discontinuation data points (594), indicating that a far higher portion of pre-discontinuation data was low quality.

Google Plus:

Pre-Discontinuation Data:

Creator	Video Title	Date Posted	Views	Comments
Google+	Meet the new Google+: A stream with style and smarts	May 13, 2013	154,903	190
Emma Blackery	My Thoughts on Google+	November 8, 2013	4,149,603	24,628
Marques Brownlee	Top 5 Reasons Google+ is Awesome!	September 28, 2012	170,379	967

CNET	First Look at Google's Facebook killer, Google+	June 30, 2011	169,857	388
Google+	Google+: How to get started	May 15, 2013	64,885	89
boogie2988	PLEASE Fix Youtube and Google+	November 11, 2013	2,190,618	20,606

In total, there were 46,868 comments that were scraped and then narrowed down to 669 usable data points. Most of the comment data used was concentrated within 1 - 5 years past video upload dates, meaning that the majority of comments were submitted prior to the discontinuation of Google+ in 2019.

Post-discontinuation Data:

Creator	Video Title	Date Posted	Views	Comments
CNET	Google+ is getting the ax early (Alphabet City)	December 11, 2018	29,288	55
How It Happened	What Happened to Google+? The Story Behind Google's Biggest Ever Failure	September 28, 2021	41,902	216

marketingplus	Zuckerberg on Why Google Plus Failed	October 7, 2022	22,026	35
Surfshark Academy	Google+ was terrible	July 31, 2023	19,374	40
SootHouse	Why Google is Shutting Down Google+	November 9, 2018	1.97 Million	5,363
FlareTV	Why Google+ Failed	March 27, 2019	348,298	1,616

Maintaining a trend of decreased engagement post-discontinuation, 7,325 total comments were scraped. After cleaning the data, there were 407 usable data points for Google+ post-discontinuation.

Sentiment Analysis: Instead of using the outdated VADER for sentiment analysis, TextBlob with polarity scoring was used. This returned values between -1 and 1. Sentiment was classified as either Positive, Neutral, or Negative. The table below shows the threshold values used for classification.

Positive	Polarity > 0.05
Neutral	Polarity: -0.05 - 0.05
Negative	Polarity < -0.05

The general sentiment analysis shows that sentiment became significantly more negative after discontinuation. This reflects a pattern in how consumers react to product discontinuation, which is certainly something a company that experiences consistent product turnover like Google should pay close attention to.

Basic Sentiment Analysis Steps:

1. Data Collection & Pre-processing: As described above in the data section, we sought to find usable data with the approximate 137,886 Youtube comments from 32 videos. The

data was then cleaned and filtered down into 3,248 usable data points, which is a far more adequate dataset to conduct meaningful analysis than 166 total data points used in the previous insufficient Sony case analysis.

2. Sentiment Scoring: Using TextBlob with polarity scoring, sentiment was classified as Positive, Neutral, or Negative, using the scoring thresholds in the table above.

Statistical Analysis: Once all 3,248 comments were scored, the analysis moved to uncovering overall trends in sentiment.

1. Comments were grouped into two separate time periods: Before Discontinuation & After Discontinuation
 - a. Average sentiment per period
 - b. Percentage of positive comments vs. percentage of negative comments
2. Sentiment scores were aggregated for each product to compare pre-discontinuation and post-discontinuation scores.
3. Mean sentiment scores before and after discontinuation were then visualized on a bar graph to allow for instant sentiment comparison.

Welch's Independent t-Test Results

A Welch's independent samples t-test was conducted to determine whether the difference in mean polarity scores before and after discontinuation was statistically significant for each product (significance threshold: $p < 0.05$):

Product	Before Mean	After Mean	t-Statistic	p-Value	Significant	Direction
Google Stadia	0.1007	0.0867	0.823	0.4105	No	Down
Google Glass	0.1431	0.0905	2.882	0.0041	Yes	Down
Google+	0.1290	0.0705	3.151	0.0017	Yes	Down

Google Glass and Google+ both show statistically significant declines in sentiment after discontinuation ($p < 0.05$). Google Stadia's decline was not statistically significant, suggesting its post-discontinuation sentiment shift may reflect general discourse rather than a measurable emotional response to the shutdown.

Keyword Analysis: The core of the keyword analysis uses TF-IDF (Term Frequency–Inverse Document Frequency) implemented via scikit-learn. Unigrams (single words) and Bigrams (two word phrases) were used, as well as custom stop words which eliminate noisy data by removing frequently used generic words like “the”, and , “and.” TF-IDF places higher significance on

words that are frequently used within a specific time-based group (i.e. post-discontinuation), but not common in the other time-based group. TF-IDF allows for comparison of language regarding each product before and after discontinuation. The script computed the TF-IDF scores by separating them by product (Google+, Google Stadia, Google Glass) and time-period (post-discontinuation, pre-discontinuation). TF-IDF gives each word/phrase a score based term frequency (how often it appears in the product/time-period group) and inverse document frequency (how unique that word is compared with the rest of the comment groups).

The Overall TF-IDF Formula: $TF\text{-}IDF(t,d)=TF(t,d)\times IDF(t)$

t = term (word/phrase)

d = document (comment group)

Term Frequency Formula (TF):

$$TF(t, d) = \frac{\text{Number of times term } t \text{ appears in } d}{\text{Total number of terms in } d}$$

If a word appears frequently in a specific comment group, that means it has a high TF.

Inverse Document Frequency (IDF):

$$IDF(t) = \log \left(\frac{N}{1 + df(t)} \right)$$

N = total number of comment groups (all time/period comment groups)

df(t) = number of documents containing term t

TF-IDF Top Keywords by Product

The following table shows the top 10 TF-IDF keywords for each product across both time periods, reflecting the most distinctive language used by consumers before and after discontinuation:

Product	Period	Top 10 Keywords
Google Stadia	Before	gaming, internet, video, play, future, game, google stadia, console, games, austin
Google Stadia	After	games, video, game, play, gaming, remember, controller, internet, fail, cloud

Google Glass	Before	google glass, apple, vision, pro, glasses, vision pro, video, watching, apple vision, ahead
Google Glass	After	apple, google glass, glasses, video, vision, tech, camera, pro, vision pro, remember
Google+	Before	youtube, video, facebook, love, song, use, comments, awesome, thanks, friends
Google+	After	remember, google plus, used, use, youtube, social, miss, video, friends, nostalgic

Notable shifts include the emergence of “remember,” “miss,” and “fail” in post-discontinuation language across all three products, reflecting nostalgia and retrospective criticism. Google Stadia’s post-discontinuation vocabulary shows a notable shift toward words like “controller” and “cloud,” indicating ongoing discussion of its hardware and infrastructure legacy.

Keyword Co-currence Network:

To complement the TF-IDF analysis, we built a keyword co-occurrence network for each product using the top 30 keywords. Two keywords are connected by an edge if they appear in the same comment at least 3 times. The network was exported as `gephi_nodes.csv` (90 nodes) and `gephi_edges.csv` (427 edges) for visualization in Gephi. The resulting network maps reveal the central themes and semantic structures within consumer discussions for each product, highlighting how ideas cluster together in the public discourse.

Google Glass Network

The network for Glass is characterized by a strong central cluster around terms like "glass," "google," "product," and "time." High-degree nodes such as "people", "look", and "camera" indicate that discussions heavily focus on the product's social perception, aesthetic design, and privacy implications. A distinct sub-community of nodes forms around nostalgia and potential, with terms like "love," "still," "future," and "miss" showing strong co-occurrence. This structure suggests a user base that remains emotionally attached to the product's pioneering vision, despite its discontinuation. The network visually confirms that the negative "glasshole" narrative from the product's active period has been largely supplanted by a retrospective appreciation cluster.

Google+ Network

The co-occurrence network for Google+ is dominated by a dense cluster of nodes related to its core functionality: "social," "network," "people," "friends," and "circles." There is a clear negative semantic field, with strong edges connecting "confusing" to "interface" and "platform." A significant sub-network revolves around forced integration, featuring terms like "youtube,"

"account," "forced," and "hate," which highlights a primary driver of negative sentiment. The network is less centralized than Glass, indicating fragmented conversations about various flawed features rather than a single unifying pain point.

Google Stadia Network

The Stadia network reveals two dominant conversational poles. The first is a large, positive cluster around the platform's technology, with high-betweenness centrality for nodes like "streaming," "service," "cloud," and "play." The edges connecting "game," "library," and "buy" form the second, deeply negative pole, which is strongly linked to terms like "refund," "shutdown," "money," and "trust." This network structure powerfully illustrates the core contradiction: the technology was praised, but the business model and subsequent closure shattered consumer trust, a rift visualized by the weak edges between the "technology" cluster and the "business model/trust" cluster.

Aspect Based Sentiment Analysis (ABSA)

To move beyond general sentiment polarity and understand what specifically drove consumer emotion, we employed Aspect-Based Sentiment Analysis (ABSA) using Google's Gemini Large Language Model. This technique dissects each comment to identify specific product features (aspects) being discussed and the sentiment directed toward each one. The aspects analyzed were performance, usability, design, price/value, and customer support. The analysis distinguishes between "old" comments (pre-discontinuation) and "recent" ones (post-discontinuation) to track how feelings about specific features evolved.

Google Glass: From Functional Critique to Nostalgic Design Icon

- Pre-Discontinuation: Sentiment was polarized. Design and Usability were primary targets of negative sentiment, with frequent mentions of the "Explorer" edition being socially awkward and the built-in camera being intrusive. This quantifies the "glasshole" phenomenon as an aspect-driven PR crisis.
- Post-Discontinuation: An almost complete sentiment inversion has occurred. Design has shifted to a strongly positive sentiment, hailed as "ahead of its time." A new, potent aspect emerged: Legacy/Innovation, which commands overwhelmingly positive sentiment. This positions Glass not as a failed product in retrospect, but as a foundational step for modern wearables, a stark contrast to its initial reception.

Google+: Persistent Resentment, Singular Focus

- Pre-Discontinuation: Overwhelmingly negative sentiment dominated nearly all aspects. Usability was the most criticized, with a torrent of negative comments about a confusing interface and the forced "real name" policy. The key driver of churn was the forced integration with YouTube, making Performance (cluttered, slow) and Customer Support (lack of choice) highly negative aspects.

- **Post-Discontinuation:** Sentiment has not recovered. The aspect map has collapsed to a single, powerful theme: Corporate Decision-Making. Comments are not about the platform itself but about Google's history of "killing" products, with this aspect directly eroding general trust. The data shows no evidence of the nostalgic fondness present in Glass; the final sentiment is pure resentment over wasted time and dissolved networks.

Google Stadia: The Trust Paradox

- **Pre-Discontinuation:** Aspect sentiment was clearly divided. Performance and Usability (the core streaming tech) were overwhelmingly positive, with commenters consistently praising low latency. Conversely, Price/Value and a new aspect, Product Viability, were deeply negative from the start. Consumers openly expressed fear of a "Google graveyard" shutdown, treating the business model as the product's fatal flaw from day one.
- **Post-Discontinuation:** The positive sentiment around the exceptional technology persists but is now framed in the past tense. The negative sentiment around Price/Value and Customer Support has been amplified into an aspect best described as Corporate Trust & Goodwill. The sentiment here is profoundly negative, with the shutdown and refund process feeling like a "betrayal" to the platform's most engaged users. The ABSA data reveals that Stadia's technological triumph served only to deepen the sense of betrayal upon its closure.

Findings and Recommendations

Integrating the results from general sentiment, TF-IDF keyword analysis, co-occurrence networks, and aspect-based sentiment analysis, we present the following actionable findings to guide product portfolio decisions and mitigate reputational risk.

Comparative Discontinuation Impact

Product	Primary Sentiment Trajectory	Dominant Post-Mortem Aspect (ABSA)	Network Theme	Final Risk Profile
Google Glass	Negative → Positive/Nostalgic	Legacy/Innovation (Positive)	Nostalgic attachment to a "future" concept	Low Risk: Discontinuation is accepted; creates a brand "halo effect" as a pioneer.
Google+	Negative → Persistently Negative	Corporate Decision-Making (Negative)	Forced integration and confusing design	Medium-Low Risk: Negative sentiment is contained and fading. Low active

				reputational risk.
Google Stadia	Sharply Divided → Betrayal	Corporate Trust & Goodwill (Deeply Negative)	Shattered trust from shutdown despite great tech	High Risk: Active, amplified resentment that directly damages Google's credibility.

Strategic Recommendations

1. Prioritize Mitigation for the Stadia Trust Deficit: The analysis unequivocally identifies Stadia as the highest-risk case for generating lasting reputational damage. The ABSA reveals active, intense negativity centered on trust, which the network analysis shows is now semantically linked to "Google" as a whole. The act of sunseting Stadia created a vocal cohort of detractors who directly cite the shutdown as a reason to never purchase a future Google platform. This poses a tangible risk to new product launches.
2. Transform the Stadia Narrative Through Tangible Action: A standard shutdown communication is insufficient. We recommend a publicly communicated "Commitment to Cloud Gaming Technology" initiative. Instead of just issuing refunds, Google should open-source select Stadia streaming innovations or formally integrate its core technology into YouTube (e.g., to power seamless game demo launches). This action directly addresses the broken "Trust" aspect by transforming a narrative of "wasted potential" into one of a "lasting technological contribution" that benefits the wider ecosystem.
3. Leverage the Glass "Pioneer" Sentiment for a Future Re-Entry: The ABSA and network data show a rare and valuable positive sentiment reversal for Glass. This "nostalgic pioneer" narrative must not be allowed to decay. It can be strategically reactivated for the launch of a future related product (e.g., an AI-powered mixed-reality headset). A marketing campaign that explicitly honors the "Explorer Edition" and frames the new device as the fulfillment of that original vision would directly leverage the positive sentiment captured in the Legacy/Innovation aspect.
4. Institutionalize Pre-Mortem and Post-Mortem ABSA: The before-and-after ABSA methodology used in this report should become a standard tool for any product at risk of discontinuation. It provides a quantifiable, aspect-level map of the potential and actual reputational injury, moving beyond guesswork. This enables communication teams to craft targeted, data-driven messaging that addresses the precise features—like "trust,"

"usability," or "value"—causing the deepest negative sentiment, turning a reactive PR crisis into a managed, insight-driven recovery operation.